NONMANDATORY APPENDIX K
(22) # NUT FACTOR CALCULATION OF TARGET TORQUE

K-1 COMMON TARGET TORQUE FORMULA

A common method for calculating target torque is to use the nut factor:

(SI Units)

$$T = KDF/1\,000 \tag{K-1M}$$

(U.S. Customary Units)

$$T = KDF/12 \tag{K-1}$$

where

D = nominal diameter of the bolt, mm (in.)
F = target bolt load, N (lb)
K = nut factor (see para. K-1.1)
T = target torque, N·m (ft-lb)

K-1.1 Nut Factor, *K*

The nut factor, K, is an experimentally determined dimensionless constant related to the coefficient of friction. The value of K in most applications at ambient temperature is considered to be approximately equal to the coefficient of friction plus 0.04 (ref. [1]) (e.g., coefficient of friction = 0.16; nut factor = 0.16 + 0.04 ~ 0.20). Published tables of experimental nut factors are available from several sources; however, care should be taken to understand the factors for the application being considered. Typical nut factors for industrial pressure vessel and piping applications using ASME SA-193 low-alloy steel bolts range from 0.16 to 0.23 at ambient temperature.

K-1.2 Effects of Changes in Nut Factor

It is important to understand the sensitivity of the obtained load to an applied torque from changes in nut factor. For example, a small change in nut factor from 0.10 to 0.30 does not result in a 20% change in torque but a 200% change.

Insufficient application of lubricant to the working surfaces adds significant variability to the obtained boltload. Research has shown that the nut factor is dependent on lubrication, bolt material, bolt diameter, bolt and nut coating, and assembly temperature. In tests using one lubricant at ambient temperature [0°C to 40°C (32°F to 100°F)], the nut factor was found to vary by 50% (from 0.155 to 0.105). In addition, in material tests, ASME SA-193 B8M bolts have been found to have a 30% higher nut factor than ASME SA-193 B7 bolts. These variables are significant and should not be ignored when selecting the lubricant and determining the nut factor (refs. [2]–[5]).

A lubricant in combination with stud material and coatings will have a specific nut factor. A change to the fastening system will change the nut factor, and the torque values will need to be adjusted accordingly. Users should use test results from nut factor trials that are similar to their own conditions (lubrication, bolt material, bolt diameter, bolt and nut coating, assembly temperature) or conduct their own nut factor trials. Nut factor trials can be conducted by applying torque to a bolt and measuring the obtained bolt load using a calibrated load cell or instrumented bolt or calibrated ultrasonic measurement.

The manufacturer's maximum temperature for a given lubrication product has not been a reliable indicator that the product improves the joint's disassembly after operation at an elevated temperature. Typically, the maximum temperature is listed as the melting point or degradation point of the solid with the highest temperature in the lubricant and is not a reflection of how the lubricant works at that higher temperature. Users should obtain test results on similar materials and in similar operating conditions to guide them in selecting the appropriate product for that service.

K-2 ADDITIONAL INFORMATION ON TARGET TORQUE FORMULAS

Additional information on torque formulas and the effect of friction factors may be found in the *Handbook of Bolts and Bolted Joints* (ref. [1]). Chapter 3 provides detailed formulas. Chapters 12 and 32 provide substantial additional theoretical and experimental information and equations, including eq. (K-2) shown in this Appendix. Equation (K-2) applies to standard 60-deg thread angle fasteners. It reflects the three specific resistance components: the thread pitch, the thread coefficient of friction, and the nut face coefficient of friction. This approach has been used in EN 1591-1, ISO 27509, and VDI 2230. Long experience has shown the nut factor method to be equally effective as the more complex formulas. While the nut factor method does not address all of the torque-preload relationship variables, it produces similar and fully acceptable values for the assembly of flanges.

For completeness, the mathematical model is given here:

$$T = \frac{F}{2}\left(\frac{p}{\pi} + \frac{\mu_t d_2}{\cos\beta} + D_e\mu_n\right) \quad \text{(K-2)}$$

This can be simplified for metric and Unified thread forms to

$$T = F\left(0.15915p + 0.57735\mu_t d_2 + \frac{D_e\mu_n}{2}\right)$$

or more approximately (from VDI 2230) to

$$T = F\left(0.16p + 0.58\mu_t d_2 + \frac{D_e\mu_n}{2}\right)$$

NOTE:

- $0.16p$ is the torque to stretch the bolt.
- $0.58\mu_t d_2$ is the torque to overcome thread friction.
- $\frac{D_e\mu_n}{2}$ is the torque to overcome face friction.

where

D_e = effective bearing diameter of the nut face, mm (in.)
= $(d_o + d_i)/2$
d_2 = basic pitch diameter of the thread, mm (in.) (For metric threads, $d_2 = d - 0.6495p$; for inch threads, $d_2 = d - 0.6495/n$.)
d_i = inner bearing diameter of the nut face, mm (in.)
d_o = outer bearing diameter of the nut face, mm (in.)
F = target bolt load, N (lb)
n = number of threads per inch, in.$^{-1}$ (applies to inch threads)
p = pitch of the thread, mm (For inch threads, this is normally quoted as threads per inch, n; i.e., $p = 1/n$.)
T = target torque, N·mm (in.-lb)
β = half included angle for the threads, deg (i.e., 30 deg for metric and Unified threads)
μ_n = coefficient of friction for the nut face or bolt head
μ_t = coefficient of friction for the threads

The target bolt load, F, can be determined from

$$F = A_s\sigma_y P_{\%}$$

where

A_s = tensile stress area of the thread, mm^2 (in.^{2}) (see Nonmandatory Appendix H)
$P_{\%}$ = percentage utilization factor for material yield strength (default value typically 50%; i.e., $P_{\%} = 0.5$)
σ_y = minimum yield strength of the bolt material, N/mm^2 (lb/in.^{2})

K-3 REFERENCES

[1] Bickford, J. H., *Handbook of Bolts and Bolted Joints*, Marcel Dekker, Inc., New York (1995), p. 233

[2] Brown, W., "Efficient Assembly of Bolted Joints," ASME 2004 Pressure Vessels and Piping Conference, PVP2004-2635, San Diego, CA, July 25–29, 2004, DOI: 10.1115/PVP2004-2635

[3] Brown, W., Marchand, L., Evrard, A., and Reeves, D., "Effect of Bolt Size on Assembly Nut Factor," ASME 2007 Pressure Vessels and Piping Conference, PVP2007-26644, San Antonio, TX, July 22–26, 2007, DOI: 10.1115/PVP2007-26644

[4] Brown, W., and Lim, T., "The Effect of Bolt Size on the Assembly Nut Factor," ASME 2015 Pressure Vessels and Piping Conference, PVP2015-45945, Boston, MA, July 19–23, 2015, DOI: 10.1115/PVP2015-45945

[5] Brown, W., and Long, S., "Factors Influencing Nut Factor Test Results," ASME 2017 Pressure Vessels and Piping Conference, PVP2017-65506, Waikoloa, HI, July 16–20, 2017, DOI: 10.1115/ PVP2017-65506